Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] **[18EE52]**

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

CONTROL SYSTEM

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Explain with examples open loop and closed loop control systems. List merits and demerits of both. **07**

b. Draw the F-V and F-I analogous circuit for the given mechanical system. **07**

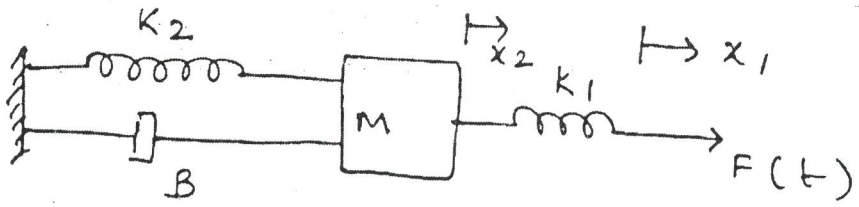

Fig 1b

2. a. State the Mason's gain formula and Define the following terms related to signal flow graph with a neat figure. **07**

- **i.** Forward Path
- **ii.** Feedback loop
- **iii.** Self loop
- **iv.** Non touching loops

b. Obtain the transfer function for the given signal flow graph. **07**

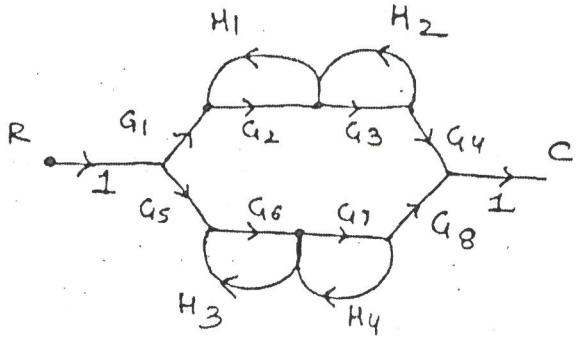

Fig 2b

3. a. Draw the transient response characteristics of a control system to a unit step input and define the following: **06**

- **i.** Delay time
- **ii.** Rise time
- **iii.** Peak time
- **iv.** Maximum overshoot;
- **v.** Settling time

b. A unity feedback control system is characterized by an open loop transfer function $G(s)H(s)= K/s(s+10)$, Determine the system gain K, so that the system will have a damping ratio of 0.5. For the value of K, find rise time, peak time, settling time and peak Overshoot. Assume that the system subjected to step of 1V **08**

OR

4. a. Sketch the root locus for the system when $G(s)H(s) = K/s(s+5)(s+10)$. **07**

b. Investigate the stability of a closed loop system whose characteristic equation is given by $S^4+2S^3+3S^2+8S+2=0$. **07**

5. a. A negative feedback control system is characterized by $G(s)=K/s(s+a)$, $H(s)=1$, find the value of K And a so that $Mr=1.04$ and $w_r=11.55$rad/sec **08**

b. List the advantages of frequency domain approach, and Explain the co-relation between time domain and frequency domain **06**

OR

6. a. Determine the value of K for a unity feedback control system having open loop transfer function **06**

$G(s)H(S)= K/s(s+2)(s+4)$, find K for GM =20dB and Find K for PM =60°.

b. Define and explain significance of the following as applied to bode plots: **08**

i. Gain margin

ii. Phase margin

iii. Gain cross over frequency

iv. phase cross over frequency

v. condition for stability of control system

7. a. Discuss the proportional controller with advantages. **08**

b. Write the difference between PI and PID controller. **06**

OR

8. a. What does a compensator do in control systems and mention necessity of compensators? **07**

b. Explains RC Lag –Lead compensator in detail with circuit Diagram. **07**